

An Intrinsic Normalization Audit for the Physical Quartic Divisor

Eight Predeclared Levels and a Sixteen-Level Frozen Atlas

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Abstract

The divisor-first pivot of RH-210–RH-211 leaves a concrete question: can a simple intrinsic normalization turn the finite physical quartic into a contracting coefficient flow? We compare three representations on eight predeclared small-noise levels and both physical channels: the raw monic quartic, determinant-radius normalization, and centered root-mean-square normalization. No parameters are fitted level by level.

The answer is negative for the two proposed normalizations. With adjacent error measured relative to the fine coefficient vector, the raw flow has range 0.04419–0.18945 and mean 0.10661. Determinant-radius normalization worsens these values to 0.06062–0.36787 and 0.19322; centered-RMS normalization gives 0.05941–0.23389 and 0.16592. Maximum left/right relative discrepancies are respectively 0.008112, 0.019881, and 0.009368. Thus the raw coefficients remain the most stable of the three tested representations.

We also freeze a sixteen-level dual-channel root atlas down to $\sigma = 0.00125$. Sparse modulus-cloud extraction exactly reproduces the previous dense-spectrum anchors while avoiding repeated dense eigensolves. The negative contraction result is useful: centered-RMS normalization exposes an exact lower-dimensional shape structure, developed in RH-213, rather than another fitted scalar law. No coefficient limit, growing determinant, Fredholm realization, or Gate-A closure is claimed.

1 Question inherited from the transport frontier

RH-207 identified a coherent left/right quartic divisor, while RH-202–RH-210 showed that naive state-space transport is not its stable realization [1–3]. Let

$$D_{\sigma,s}(z) = \prod_{j=1}^4 (z - \lambda_{\sigma,s,j}) = z^4 + c_{1,\sigma,s}z^3 + c_{2,\sigma,s}z^2 + c_{3,\sigma,s}z + c_{4,\sigma,s}, \quad (1)$$

where $s \in \{L, R\}$ labels the two physical channels. Three coarse anchors showed channel coherence but not stationarity in σ .

The present audit asks a deliberately narrow question. Is the scale motion mostly an affine gauge that can be removed by one natural formula at every level? A positive answer

would produce a candidate compact coefficient family. A negative answer rules out only the tested formulas, not every renormalization.

2 Predeclared scale protocol

The normalization comparison uses

$$0.04, 0.032, 0.025, 0.02, 0.016, 0.0125, 0.01, 0.008. \quad (2)$$

The frozen atlas additionally contains

$$0.00625, 0.005, 0.004, 0.0032, 0.0025, 0.002, 0.0016, 0.00125. \quad (3)$$

The last eight points are not used to redefine the RH-212 winner. They are published so that subsequent papers can state their own training and holdout rules without recomputing the matrix spectrum.

At each scale we retain the same inherited physical rule: remove the Perron and negative parity modes and select the next four eigenvalues by modulus. The selected set is checked for conjugate closure. All 32 endpoint packets are conjugate closed to the reported floating precision.

3 Three intrinsic representations

3.1 Raw monic coefficients

The baseline is simply the coefficient vector

$$C_{\sigma,s}^{\text{raw}} = (1, c_1, c_2, c_3, c_4). \quad (4)$$

It retains location, radius, and shape. It is similarity invariant but not invariant under affine changes of the spectral variable.

3.2 Determinant-radius normalization

Set

$$\rho = \left| \prod_{j=1}^4 \lambda_j \right|^{1/4} = |c_4|^{1/4}, \quad \hat{\lambda}_j = \lambda_j / \rho. \quad (5)$$

Then the constant coefficient of $\prod_j (z - \hat{\lambda}_j)$ has modulus one.

Proposition 3.1. *For every quartet with nonzero product, ρ is invariant under root permutation and operator similarity. The normalized constant term has modulus one.*

Proof. Both statements follow from the elementary symmetric product and $\prod_j \hat{\lambda}_j = \rho^{-4} \prod_j \lambda_j$. $\square$

This normalization does not remove translation. It can also amplify the remaining coefficients when $|c_4|$ is small.

3.3 Centered RMS normalization

Define

$$\mu = \frac{1}{4} \sum_{j=1}^4 \lambda_j, \quad r = \left(\frac{1}{4} \sum_{j=1}^4 |\lambda_j - \mu|^2 \right)^{1/2}, \quad q_j = \frac{\lambda_j - \mu}{r}. \quad (6)$$

Proposition 3.2. *For every nonconstant quartet,*

$$\frac{1}{4} \sum_j q_j = 0, \quad \frac{1}{4} \sum_j |q_j|^2 = 1. \quad (7)$$

The multiset $\{q_j\}$ is invariant under translation $\lambda_j \mapsto \lambda_j + a$ and positive dilation $\lambda_j \mapsto b\lambda_j$ with $b > 0$.

Proof. Centering gives the first identity and the definition of r gives the second. Under translation, the barycenter translates by the same amount. Under positive dilation, both centered roots and r are multiplied by b . $\square$

For conjugate-closed quartets, μ is real and the normalized polynomial has real coefficients and zero cubic coefficient. This exact structural gain will matter even though contraction fails.

4 Error functional

For adjacent coarse/fine scales $\sigma_c > \sigma_f$, write C_c, C_f for one of the three monic coefficient vectors and define

$$E_f(C_c, C_f) = \frac{\|C_f - C_c\|_2}{\|C_f\|_2}. \quad (8)$$

The fine denominator is fixed in advance and treats the refined endpoint as the target. Maximum absolute coefficient error and the reverse denominator are also archived, but no conclusion depends on changing the metric after inspection.

At a fixed scale, channel discrepancy is

$$E_{LR} = \frac{\|C_L - C_R\|_2}{\|C_L\|_2}. \quad (9)$$

5 Finite normalization verdict

The 14 adjacent/channel cases for each representation give:

representation	min E_f	max E_f	mean E_f	max E_{LR}
raw	0.04419	0.18945	0.10661	0.008112
determinant radius	0.06062	0.36787	0.19322	0.019881
centered RMS	0.05941	0.23389	0.16592	0.009368

Proposition 5.1 (Finite negative result). *On the eight predeclared levels and both channels, neither determinant-radius nor centered-RMS normalization improves the mean, maximum, or minimum adjacent fine-relative coefficient error simultaneously. The raw vector has the smallest mean and maximum, and it also has the smallest worst channel discrepancy.*

Proof. This is the direct comparison in the displayed table, generated from the frozen endpoint roots by the exact formulas above. $\square$

The word “negative” is important. The data do not establish that every intrinsic normalization fails. They reject two concrete formulas as a uniformly better cross-scale coordinate on this audit.

6 Why normalization can worsen the flow

The determinant radius forces one coefficient to a fixed modulus, but the other coefficients transform with different powers of ρ^{-1} . Small relative motion in c_4 can therefore induce larger motion in lower-order terms. The operation removes one scalar without separating translation from shape.

Centered-RMS normalization removes two affine degrees of freedom. If the remaining shape itself moves substantially, the normalized vector must show that motion more clearly. Consequently a larger coefficient error is not a defect of the quotient. It can be evidence that location and radius were masking a genuine shape trajectory.

Indeed, every centered-RMS polynomial in the atlas has zero cubic term and lies, to roundoff, on one algebraic surface in the remaining three coefficients. RH-213 proves that this surface is exactly two-dimensional. Thus RH-212 changes the next question from “which scalar should be fitted?” to “what exact geometry remains after the affine gauge is removed?”

7 Sparse extraction and reproducibility

The historical construction formed a dense deflated bulk operator. Its outer nontrivial eigenvalues are unchanged if one instead computes a modulus cloud of the original sparse folded-Gaussian matrix, removes the Perron and negative real parity roots, and takes the next quartet. At the old anchors 0.04, 0.02, 0.01, the resulting roots and coefficients reproduce the frozen dense results to floating roundoff.

This shortcut is not a new spectral theorem. It is an algebraic consequence of replacing two selected eigenvalues by zero in the inherited finite model, combined with a cloud large enough to resolve the radial gap. The archive retains dimensions, roots, coefficients, centers, radii, and conjugacy diagnostics for every endpoint.

8 Claim boundary and route consequence

The following remain open:

1. existence of either a raw or normalized coefficient limit;
2. stability under all smaller noise levels;
3. a rank-growing cloud with controlled omitted factors;
4. local uniform convergence to a nonzero analytic determinant;

5. a dynamical or Fredholm realization.

Gate A is therefore open. Gates B–E are untouched. No self-adjoint operator, $T \log T$ law, arithmetic trace identity, zeta divisor, or RH statement is inferred. The immediate route is the exact centered shape manifold, not another post-hoc normalization.

References

- [1] Liang Wang. Dual-channel coherence of the quartic spectral divisor, 2026. RH-207 preprint.
- [2] Liang Wang. A divisor-first pivot inside gate a, 2026. RH-210 preprint.
- [3] Liang Wang. Ten layers at the cross-scale transport frontier, 2026. RH-211 preprint.